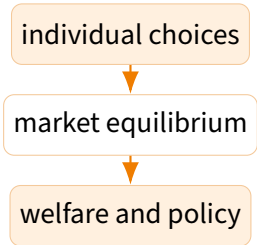
Unit 4: Markets and Welfare

How prices coordinate choices

- competitive equilibrium and gains from trade;
- taxes, price controls, and market power;
- externalities and corrective policy;
- general equilibrium and the welfare theorems.



The Price Was Given. Now We Explain It

Unit 1: Consumers

At each price, consumers choose the best affordable bundle. Their choices determine demand.

$$D(p) = \sum_i x_i^*(p).$$

Unit 3: Firms

At each price, firms choose inputs and output to maximize profit. Their choices determine supply.

$$S(p) = \sum_j q_j^*(p).$$

The price must make these independently chosen quantities compatible.

Unit 4 asks which price clears the market, who gains from trade, and how the outcome changes under a different policy or market structure.

A Competitive Market

- Buyers and sellers trade the same good at a common price.
- Each participant takes that price as given when choosing a quantity.
- Buyers can compare offers. Sellers compete for buyers.
- The market price adjusts so that total demand equals total supply.

Price taking is a behavioral assumption. Many small participants trading a homogeneous good make it a plausible approximation.

Free entry is an additional assumption when we study long-run adjustment.

Competitive Equilibrium

Equilibrium

A price p^* and quantities such that every consumer and firm optimizes at p^*, and the market clears:

$$\boxed{D(p^*) = S(p^*) = Q^*.}$$

- At too low a price, desired purchases exceed desired sales.
- At too high a price, desired sales exceed desired purchases.
- At p^*, the choices on both sides are compatible.

Market clearing is an equilibrium condition. A separate model would be needed to describe the exact process of price adjustment.

Partial Equilibrium: One Market at a Time

We study the price and quantity of one good, holding the surrounding conditions fixed.

Demand conditions	Supply conditions
Consumer incomes	Input prices
Preferences	Technology
Prices of other goods	Installed capacity and the set of firms

A policy can change one of these conditions. We then solve for a new equilibrium.

Partial equilibrium is useful when feedback through other markets is small. General equilibrium later allows those prices and incomes to adjust together.

One Market We Will Follow

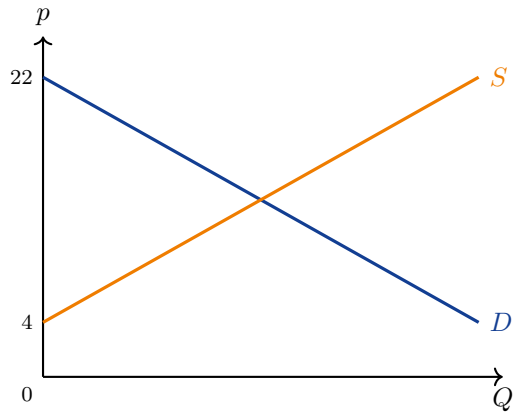
For the relevant range of prices,

$$D(p) = 22 - p, \qquad S(p) = p - 4.$$

The inverse curves are

$$P_D(Q) = 22 - Q, \qquad P_S(Q) = 4 + Q.$$

- Q: units traded per week.
- p: euros per unit.
- Demand is zero above 22.
- Supply is zero below 4.



All numbers are illustrative. The same preferences and technology will anchor our policy comparisons.

Solving for Price and Quantity

Set desired purchases equal to desired sales:

$$22 - p = p - 4.$$

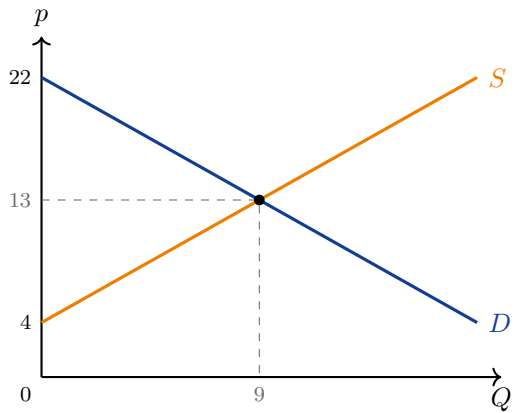
Therefore,

$$\boxed{p^* = 13, \qquad Q^* = 9.}$$

Check both sides:

$$D(13) = 9, \qquad S(13) = 9.$$

The equilibrium price is an outcome of the whole market.



Excess Demand and Excess Supply

Price	$D(p)$	$S(p)$	$D(p) - S(p)$	Market imbalance
10	12	6	6	Excess demand
13	9	9	0	Market clears
16	6	12	-6	Excess supply

- At $p = 10$, some buyers cannot obtain their desired quantity.
- At $p = 16$, some sellers cannot sell their desired quantity.

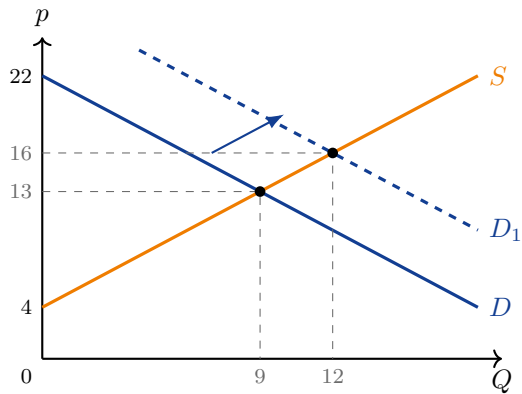
Desired quantities can differ. Actual purchases and actual sales are always equal because each transaction has both a buyer and a seller.

A Demand Increase

Suppose demand rises to $D_1(p) = 28 - p$ while supply stays $S(p) = p - 4$.

$$28 - p = p - 4 \quad \Rightarrow \quad p_1 = 16,\ Q_1 = 12.$$

- At each price, buyers want more.
- The higher market price induces firms to expand production.



Demand shifts. The equilibrium moves along the unchanged supply curve.

A Cost Increase

Suppose marginal costs rise by 6 at every quantity:

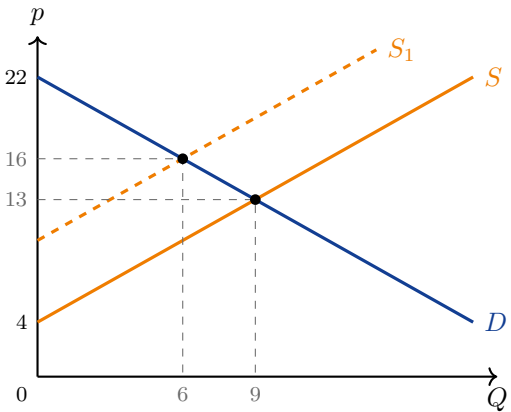
$$P_{S,1}(Q) = 10 + Q.$$

Demand remains $P_D(Q) = 22 - Q$.

$$22 - Q = 10 + Q.$$

$$Q_1 = 6, \qquad p_1 = 16.$$

Prices rise while production falls.



A higher observed price can reflect stronger demand or higher costs. The quantity response helps distinguish the two.

Observed Data Are Equilibrium Outcomes

Demand varies, supply stays fixed

Prices and quantities move together along supply. The observed relationship can slope upward even though demand slopes downward.

Supply varies, demand stays fixed

Prices and quantities move in opposite directions along demand. This variation can reveal the demand response if the demand conditions really stay fixed.

When both curves shift, the scatter of market outcomes generally traces neither curve.

Data-science lesson

A regression of quantity on price does not automatically estimate demand. We need variation that shifts supply without directly shifting demand, or another credible identification strategy.

Equilibrium over Time

A sustained demand increase creates different opportunities at different horizons.

Horizon	What can adjust?	Market implication
Short run	Output of existing firms	Higher demand can raise price and profits.
Long run	Plant size, entry, and exit	Profitable opportunities attract capacity and entry.

With free entry by identical firms and constant input prices,

$$p^* = \min_q \mathrm{LRAC}(q), \qquad \pi^* = 0.$$

Zero economic profit pays the owner's inputs at their opportunity costs. Scarce inputs and firm differences can sustain rents.

Exercise: A New Market Equilibrium

Demand becomes $D_1(p) = 28 - p$. Costs also rise, so supply becomes $S_1(p) = p - 8$.

(a) Solve for the new equilibrium price and quantity.

(b) Compare them with the original $p^* = 13$ and $Q^* = 9$.

(c) Could the price change alone tell us whether demand or costs changed?

Before calculating, predict which changes have an unambiguous effect on price and which have opposing effects on quantity.

Solution: Both Curves Shift

$$28 - p = p - 8 \quad \Rightarrow \quad \boxed{p_1 = 18, \quad Q_1 = 10.}$$

Relative to the original equilibrium,

$$\Delta p = 5, \qquad \Delta Q = 1.$$

- Both stronger demand and higher costs push the price up.
- Their effects on quantity oppose each other.
- In this example, the demand increase dominates for quantity.

Counterfactuals require knowing which underlying conditions changed. A price change alone cannot identify the cause.

The Gain from One Trade

A buyer is willing to pay up to **12 euros** for a unit. Producing it has an opportunity cost of **7 euros**.

Trading price	Buyer gains	Seller gains	Total gain
8	$12 - 8 = 4$	$8 - 7 = 1$	5
10	$12 - 10 = 2$	$10 - 7 = 3$	5

$$\text{Total gain} = \text{willingness to pay} - \text{opportunity cost}.$$

The price divides the gain between buyer and seller. When both sides are included, the payment is a transfer.

Demand and Willingness to Pay

Unit 1 derived demand from the consumer's optimal choice. To measure gains in euros, consider

$$u(x, y) = v(x) + y, \qquad px + y \leq m,$$

where y is other consumption with price normalized to one. If the consumer has enough income for an interior choice of y,

$$\max_x \{v(x) + m - px\} \quad \Rightarrow \quad v'(x) = p.$$

- $v'(x)$ measures marginal willingness to pay in euros.
- The inverse demand curve records this marginal value.

This quasilinear benchmark makes demand-area welfare calculations exact. With income effects, ordinary consumer-surplus changes are generally an approximation.

Demand and Marginal Willingness to Pay

A household has income 20 and preferences

$$u(x, y) = xy,$$

where y is money left for other goods, whose price is 1.

Its budget is $px + y = 20$. At an interior optimum,

$$\frac{MU_x}{MU_y} = \frac{y}{x} = p \quad \Longrightarrow \quad x(p) = \frac{10}{p}, \qquad y(p) = 10.$$

Price	Quantity purchased	Money remaining
10	1	10
5	2	10

Demand reveals marginal willingness to pay at each chosen bundle.

But these are different purchasing situations: the lower price applies to *all* units.

How Much Would You Pay for One More Unit?

Suppose the household has already bought one unit for 10. It now holds

$$(x, y) = (1, 10), \qquad u(1, 10) = 10.$$

How much would it pay for **one additional unit**, keeping its earlier purchase unchanged?

The maximum payment T leaves it indifferent:

$$u(2, 10 - T) = u(1, 10) \quad \Longrightarrow \quad 2(10 - T) = 10.$$

Hence,

$$\boxed{T = 5.}$$

- Buying both units at a uniform price of 5 leaves the bundle $(2, 10)$.
- Paying 5 for an extra unit after buying the first for 10 leaves $(2, 5)$.

The quantity is the same, but the money remaining differs.

‘Adding Up Willingness to Pay”?

Imagine moving from one unit to two in tiny increments Δx.

If $w(x)$ is marginal willingness to pay along this process, the payment for a tiny increment is approximately

$$w(x)\,\Delta x.$$

Adding these payments gives

$$w(1)\Delta x + w(1+\Delta x)\Delta x + \cdots \quad \longrightarrow \quad \int_1^2 w(x)\,dx.$$

Using ordinary inverse demand, $p(x) = 10/x$, would give

$$\int_1^2 \frac{10}{x}\,dx = 10\ln 2 \approx 6.93.$$

But we just calculated a maximum payment of 5.

Which marginal valuations should we be adding?

Paying for Each Increment

Start at $(1, 10)$. As the household receives tiny increments and pays its maximum for each, its utility stays at 10:

$$xy = 10 \quad \Longrightarrow \quad y(x) = \frac{10}{x}.$$

Its marginal willingness to pay along this path is

$$w(x) = \frac{MU_x}{MU_y} = \frac{y(x)}{x} = \frac{10}{x^2}.$$

Quantity	Ordinary inverse demand	Along the payment path
1	10	10
1.5	6.67	4.44
2	5	2.50

Adding the valuations along the payment path gives

$$\boxed{\int_1^2 \frac{10}{x^2}\, dx = 5.}$$

What Quasilinearity Simplifies

Now suppose preferences are

$$u(x, y) = 10 \ln x + y.$$

Marginal willingness to pay is

$$\frac{MU_x}{MU_y} = \frac{10}{x}.$$

It depends on x, but not on money remaining.

Starting at $(1, 10)$, the maximum payment for one more unit solves

$$10 \ln 2 + 10 - T = 10 \ln 1 + 10.$$

Therefore,

$$\boxed{T = 10 \ln 2 = \int_1^2 \frac{10}{x}\, dx \approx 6.93.}$$

Paying for earlier increments does not change marginal willingness to pay at a given quantity.

Consumer Surplus

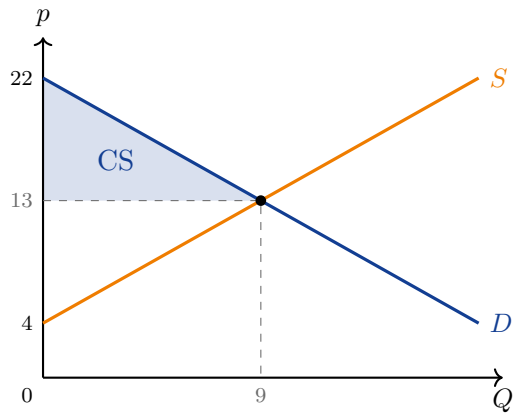
Consumer surplus is total willingness to pay minus expenditure:

$$\text{CS} = \int_0^Q P_D(z)\,dz - pQ.$$

At $p = 13$ and $Q = 9$,

$$\text{CS} = \tfrac{1}{2}(22 - 13)9 = 40.5.$$

Buyers value earlier units more than the price they pay.



Producer Surplus

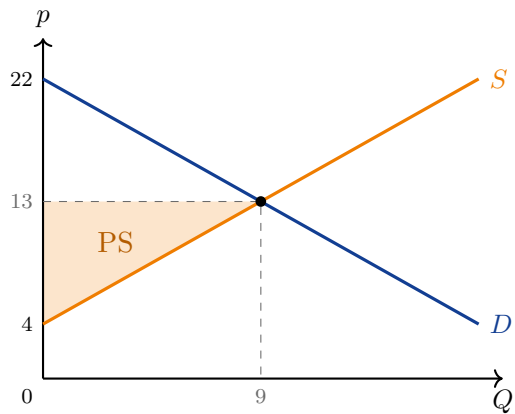
With increasing marginal costs and no avoidable setup costs, supply records the marginal cost of another unit.

$$\text{PS} = pQ - \int_0^Q P_S(z)\,dz.$$

At $p = 13$ and $Q = 9$,

$$\text{PS} = \tfrac{1}{2}(13 - 4)9 = 40.5.$$

The common price exceeds the cost of earlier units.



Producer Surplus and Profit

Recall Unit 3's short-run cost decomposition:

$$C(q) = F + VC(q), \qquad VC(0) = 0.$$

When F is unavoidable for the decision being studied,

$$\text{PS} = pq - VC(q), \qquad \pi = \text{PS} - F.$$

Revenue	100	
Variable cost	60	$\text{PS} = 40$
Unavoidable fixed cost	50	$\pi = -10$

A firm can have positive producer surplus and negative profit. Fixed costs that change with entry or exit must be included in welfare comparisons across those choices.

Total Surplus

With no externalities or government payments,

$$W(Q) = \text{CS} + \text{PS}.$$

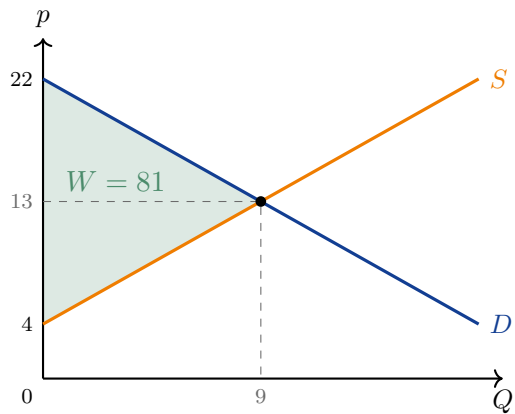
Payments cancel, giving

$$W(Q) = \int_0^Q [P_D(z) - P_S(z)]\, dz.$$

In our market,

$$W(Q) = 18Q - Q^2.$$

At $Q^* = 9$, total surplus is 81.



Which Quantity Maximizes Surplus?

$$W'(Q) = P_D(Q) - P_S(Q).$$

Quantity	Value of another unit	Welfare implication
$Q < 9$	Willingness to pay exceeds cost	More trade increases surplus.
$Q = 9$	Willingness to pay equals cost	All marginal gains are exhausted.
$Q > 9$	Cost exceeds willingness to pay	Less trade increases surplus.

The competitive quantity maximizes total surplus in our quasilinear benchmark when consumption and production are efficiently allocated.

This conclusion also requires efficient allocation across buyers and across producers.

Efficiency Also Concerns Who Trades

An event sells tickets for 6 euros by lottery. Resale is prohibited. The last ticket goes to Ana, while Bruno receives none.

	Ana	Bruno
Willingness to pay	8	12
Receives the last ticket?	Yes	No

Giving the ticket to Bruno raises its consumption value by $12 - 8 = 4$ euros. The number of tickets and the production cost stay the same.

For a given quantity, surplus is highest when units go to the highest-value buyers and production comes from the lowest-cost suppliers.

More Surplus and Who Gains

Taking Ana's ticket and giving it to Bruno raises surplus, but harms Ana. Suppose instead that Bruno can buy Ana's ticket for 10 euros.

	Keeps the ticket	Resells for 10	Change
Ana's gain	$8 - 6 = 2$	$10 - 6 = 4$	$+2$
Bruno's gain	0	$12 - 10 = 2$	$+2$

The organizer still receives 6. No one else is affected.

Pareto improvement

A change makes someone better off and no one worse off.

More surplus creates scope for compensation here. A policy need not actually compensate its losers.

How a Common Price Coordinates Choices

At a smooth interior competitive equilibrium,

$$\underbrace{v_i'(x_i)}_{\text{marginal value to buyer } i} = \underbrace{p}_{\text{common price}} = \underbrace{C_j'(q_j)}_{\text{marginal cost at firm } j} .$$

- Buyers adjust consumption until their marginal valuations agree.
- Firms adjust production until their marginal costs agree.
- Market clearing makes total purchases equal total production.

Price taking and unrestricted trade at a market-clearing price align marginal values and costs. A common posted price with rationing need not do so.

When Competition Maximizes Surplus

The result: competitive equilibrium maximizes total surplus in our quasilinear benchmark, including who consumes and who produces.

Condition	What a failure can change
Price taking	Market power can restrict output.
All relevant costs and benefits priced	Externalities can create too much or too little activity.
Information needed for trade	Hidden quality or private information can prevent valuable trades.
Access to relevant markets	Missing insurance or credit markets can constrain choices.

We also require market clearing without rationing or costly allocation.

A Per-Unit Tax Creates a Price Wedge

Let t be the tax collected per unit traded.

$$\boxed{p_b - p_s = t,}$$

where p_b is what buyers pay and p_s is what sellers keep. The equilibrium satisfies

$$D(p_b) = S(p_s) = Q_t.$$

Equivalently,

$$P_D(Q_t) = P_S(Q_t) + t.$$

The tax changes the prices relevant for each side's decision. We must solve for both prices and the traded quantity.

A Tax of Six Euros

In our market,

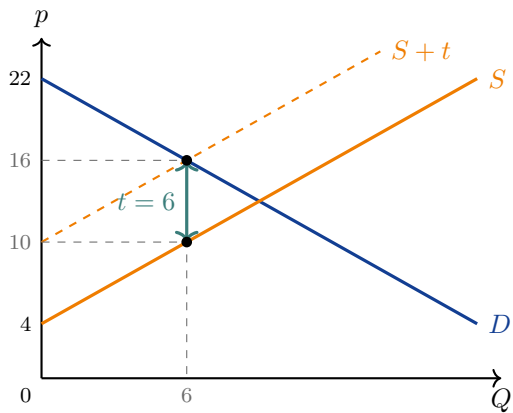
$$22 - Q_t = 4 + Q_t + 6.$$

Therefore,

$$\boxed{Q_t = 6, \quad p_b = 16, \quad p_s = 10.}$$

Relative to the original price 13,

- buyers pay 3 more;
- sellers keep 3 less.



Who Remits the Tax?

Sellers remit it

Sellers receive the buyer's payment and send t per unit to the government. They keep $p_s = p_b - t$.

Buyers remit it

Buyers pay sellers p_s and send t per unit to the government. Their total cost is $p_b = p_s + t$.

Both arrangements imply the same conditions:

$$p_b - p_s = t, \qquad D(p_b) = S(p_s).$$

Under the same enforcement and transaction costs, statutory responsibility does not determine economic incidence. Demand and supply responses do.

Elasticity Determines Tax Incidence

The side that finds it harder to adjust quantity bears more of a small tax. At an initially untaxed competitive equilibrium, define

$$\varepsilon_D = -D'(p)\frac{p}{Q} > 0, \qquad \varepsilon_S = S'(p)\frac{p}{Q} \geq 0.$$

For a small tax change,

$$\frac{dp_b}{dt} = \frac{\varepsilon_S}{\varepsilon_S + \varepsilon_D}, \qquad -\frac{dp_s}{dt} = \frac{\varepsilon_D}{\varepsilon_S + \varepsilon_D}.$$

- Nearly fixed demand: buyers bear most of the tax.
- Nearly fixed supply: sellers bear most of the tax.

Tax Revenue and Deadweight Loss

At $t = 6$,

$$\text{CS} = \tfrac{1}{2}(22 - 16)6 = 18,$$

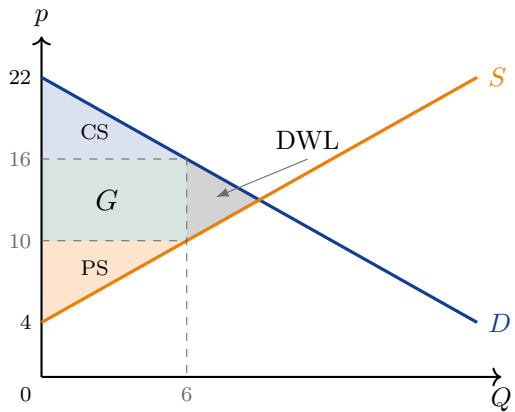
$$\text{PS} = \tfrac{1}{2}(10 - 4)6 = 18,$$

$$G = tQ_t = 36.$$

Counting tax revenue,

$$W_t = \text{CS} + \text{PS} + G = 72.$$

$$\text{DWL} = 81 - 72 = 9.$$



The Value of Lost Trades

The tax removes the trades between $Q = 6$ and $Q = 9$.

$$\text{DWL} = \int_6^9 \underbrace{[(22 - z) - (4 + z)]}_{\text{gain from another unit}} \, dz = 9.$$

For linear demand and supply,

$$\text{DWL} = \tfrac{1}{2} \times \text{tax wedge} \times \text{quantity reduction}.$$

- Tax revenue is a transfer to the government.
- Deadweight loss is surplus that no one receives.
- With locally linear curves, larger quantity responses create larger losses.

This conclusion uses a market with no pre-existing distortion or externality. A corrective tax will change the comparison.

When a Trade Affects Other People

An externality arises when an action affects someone else's welfare through a channel that the decision-maker does not fully face in market prices.

- Production can impose pollution damages on nearby residents.
- Vaccination can reduce infection risk for other people.

For a negative production externality,

$$\boxed{\mathrm{SMC}(Q) = \mathrm{MC}(Q) + \mathrm{MEC}(Q),}$$

where MEC is marginal external cost.

A competitive price can coordinate buyers and sellers while omitting costs borne by people outside the transaction.

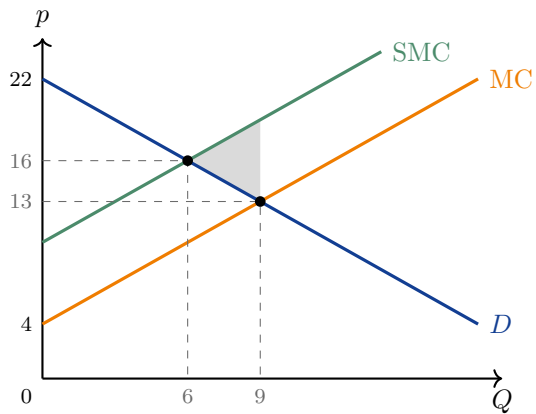
External Costs and Excess Output

Keep $P_D(Q) = 22 - Q$ and private $\mathrm{MC}(Q) = 4 + Q$. Suppose each unit causes 6 euros of external damage.

$$\mathrm{SMC}(Q) = 10 + Q.$$

Competition still selects $Q_C = 9$. Efficiency requires

$$22 - Q_E = 10 + Q_E \Rightarrow Q_E = 6.$$



A Corrective Tax Aligns Private Incentives

A per-unit tax of $t = 6$ makes producers face the external cost.

$$p_b = \mathrm{MC}(Q) + 6 = \mathrm{SMC}(Q).$$

The taxed competitive equilibrium has

$$Q = 6, \qquad p_b = 16, \qquad p_s = 10.$$

More generally, the corrective tax equals marginal external damage at the efficient quantity:

$$\boxed{t^* = \mathrm{MEC}(Q_E).}$$

The same tax that reduced welfare in a market without externalities now corrects an existing incentive problem.

Corrective Tax: Costs and Benefits

The tax reduces output from 9 to 6.
Grey: private gains from the lost trades.

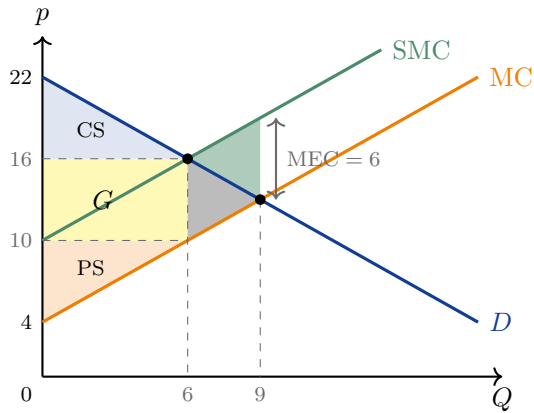
$$\tfrac{1}{2}(9-6)(16-10)=9.$$

Grey plus green: avoided external damage.

$$6(9-6)=18.$$

Green: net social gain.

$$\boxed{\Delta W = 18-9=9.}$$



Green is the deadweight loss of the untaxed outcome that the tax removes. Yellow is tax revenue, a transfer. It does not measure the damage avoided.

Welfare Must Include External Damages

Social welfare is now

$$W = \text{CS} + \text{PS} + G - \text{external damage.}$$

	No tax	Corrective tax $t = 6$
Quantity	9	6
Consumer plus producer surplus	81	36
Government revenue	0	36
External damage	54	36
Social welfare	27	36

The tax sacrifices 9 in gains from market trades and avoids 18 in damage. Net welfare rises by 9.

Returning to the Power Plants in Unit 3

Unit 3 asked how an emissions charge changes a plant's costs, output, and renewal decision. Unit 4 adds the rest of the market and the people affected by pollution.

- Lower supply can change the electricity price and consumer surplus.
- Production may shift to plants with different emissions and costs.
- Lower emissions can create benefits outside the electricity market.

Plant profits and closures alone cannot establish the welfare effect of an emissions charge. We need the market response and the external damages.

Subsidies Can Also Create Deadweight Loss

Let producers receive a subsidy $s = 6$ per unit. Then

$$p_s - p_b = 6, \qquad 22 - Q_s + 6 = 4 + Q_s.$$

Thus $Q_s = 12$, buyers pay 10, and sellers receive 16.

Consumer surplus	Producer surplus	Government balance	Total surplus
72	72	-72	72

The extra units from 9 to 12 cost more to produce than buyers value them. The subsidy induces overproduction and a loss of 9.

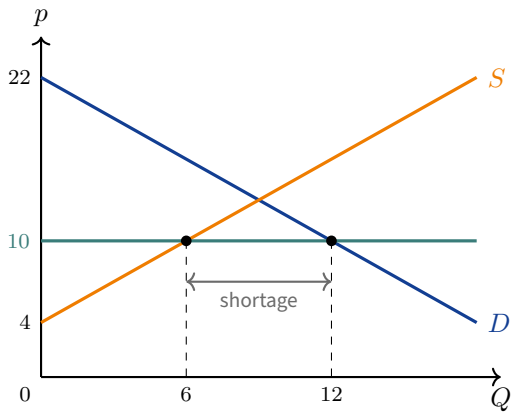
A Price Ceiling Creates a Shortage

A ceiling is a legal maximum price. Let $\bar{p} = 10 < 13$.

$$D(10) = 12, \qquad S(10) = 6.$$

- Buyers want 12 units.
- Firms supply 6 units.
- The shortage is 6 units.

If all available units trade, actual sales equal 6.



Who Gets the Six Available Units?

The six units go to the highest-value buyers, without allocation costs.

$$\mathrm{CS} = \int_0^6 [(22 - z) - 10]\, dz = 54,$$

$$\mathrm{PS} = \int_0^6 [10 - (4 + z)]\, dz = 18.$$

	Competition	Ceiling with efficient rationing
Consumer surplus	40.5	54
Producer surplus	40.5	18
Total surplus	81	72

Aggregate consumer surplus rises under efficient rationing, even though some buyers lose access.

Rationing Can Create Additional Losses

For a unit-demand illustration, rank the 12 units requested at $p = 10$ using the smooth demand curve. Their values are uniformly spread from 10 to 22.

- Efficient rationing selects the top six units, with average value 19.
- Random rationing gives each unit demand the same chance. Average value among recipients is 16.

Under random rationing, expected consumer surplus is

$$6(16 - 10) = 36, \qquad \mathrm{PS} = 18, \qquad W = 54.$$

Relative to efficient rationing, misallocation destroys another 18 of surplus. Waiting, search, or favoritism can add further costs.

A Price Floor Creates Excess Supply

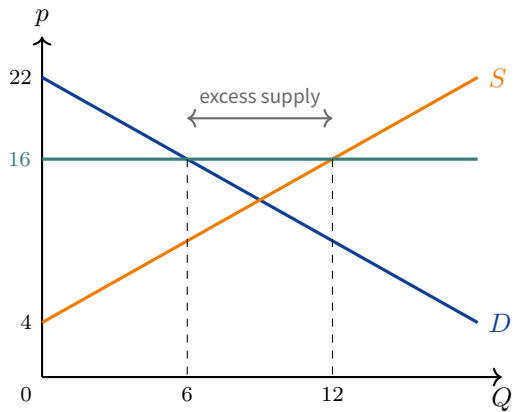
A floor is a legal minimum price. Let $\underline{p} = 16 > 13$.

$$D(16) = 6, \qquad S(16) = 12.$$

Assume the government buys nothing and the lowest-cost sellers make the six sales.

$$\text{CS} = 18, \quad \text{PS} = 54, \quad W = 72.$$

Other allocations can yield less surplus.



Price Controls and the Adjustment Horizon

Consider a ceiling on rental prices in a competitive housing benchmark.

- Existing housing makes supply relatively unresponsive in the short run.
- Over time, construction, maintenance, and conversion can respond.
- Current tenants, prospective tenants, and owners may experience different effects.

The relevant question combines the price rule, the allocation rule, and the horizon over which supply can adjust.

Data-science lesson

Measuring rents for remaining tenants alone misses households that cannot obtain a tenancy, changes in quality, and changes in supply.

Exercise: Comparing Interventions

Return to $P_D(Q) = 22 - Q$ and $P_S(Q) = 4 + Q$.

(a) Compare a tax of 6 with a ceiling of 10. Why do both produce $Q = 6$?

(b) Which policy gives buyers more surplus under efficient rationing?

(c) Where does the difference go?

(d) Would the answer survive a different ceiling allocation rule?

Separate the number of units traded from the distribution of gains.

Equal Quantities, Different Gains

Outcome	Q	CS	PS	Government	Total
Competition	9	40.5	40.5	0	81
Tax $t = 6$	6	18	18	36	72
Ceiling $\bar{p} = 10$	6	54	18	0	72
Floor $\underline{p} = 16$	6	18	54	0	72
Subsidy $s = 6$	12	72	72	-72	72

The tax and ceiling both trade six units. Under efficient rationing, the ceiling moves the tax's 36 of government revenue to buyers.

Price controls assume efficient rationing, no government purchases, and no allocation costs. The benchmark has no externalities or changing fixed costs.

When the Seller Can Affect the Price

Competitive firm

The price is given. Selling one more unit adds p to revenue.

$$\max_q \{pq - C(q)\}.$$

Single-price monopolist

The seller faces market demand. Selling more requires a lower price for all units.

$$\max_Q \{P_D(Q)Q - C(Q)\}.$$

A monopoly may persist because of entry barriers, exclusive rights, or a cost advantage.

We hold demand and the available technology fixed to isolate the effect of market power.

Marginal Revenue Includes a Price Effect

Total revenue is $R(Q) = P_D(Q)Q$. Differentiating gives

$$\boxed{\mathrm{MR}(Q) = P_D(Q) + QP_D'(Q).}$$

$P_D(Q)$	Revenue from a little more output at the current price
$QP_D'(Q)$	Revenue lost when the price falls on existing sales

With downward-sloping demand, $P_D'(Q) < 0$, so $\mathrm{MR}(Q) < P_D(Q)$ for $Q > 0$.

The consumer values the marginal unit at its price. The monopolist also counts the price reduction on units it was already selling.

The Monopoly Choice

Keep demand $P_D(Q) = 22 - Q$ and the same total variable cost:

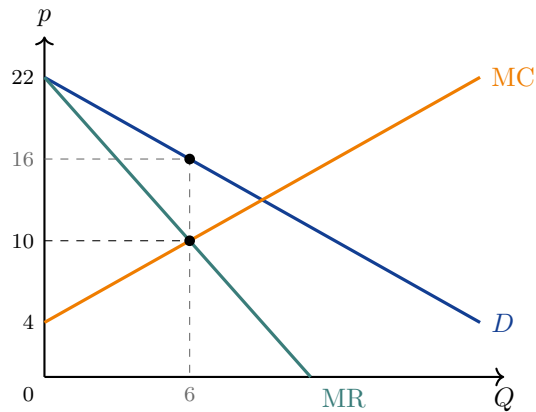
$$C(Q) = 4Q + \tfrac{1}{2}Q^2.$$

Then $\mathrm{MR} = 22 - 2Q$ and $\mathrm{MC} = 4 + Q$.

$$22 - 2Q = 4 + Q \Rightarrow Q_M = 6.$$

Read the price from demand:

$$\boxed{p_M = 16, \qquad \mathrm{MC}(6) = 10.}$$



Monopoly Welfare

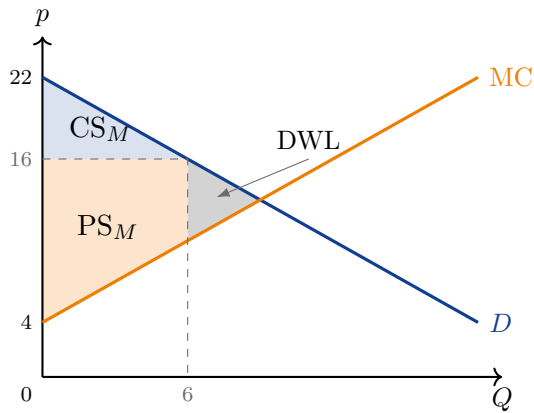
At $Q_M = 6$ and $p_M = 16$,

$$\text{CS}_M = 18, \qquad \text{PS}_M = 54,$$

$$W_M = 72, \qquad \text{DWL}_M = 9.$$

Compared with competition,

- consumers lose 22.5;
- producers gain 13.5;
- society loses 9.



The Trades a Monopolist Leaves Unmade

At the monopoly quantity,

$$P_D(6) = 16 > \mathrm{MC}(6) = 10.$$

Consumers would pay more for another unit than it costs to produce. But increasing output also lowers the price on existing sales, so

$$\mathrm{MR}(6) = 10 = \mathrm{MC}(6).$$

The monopolist stops while gains from additional trade remain. Private profit maximization and total-surplus maximization select different quantities.

The transfer from consumers to the seller is distinct from the deadweight loss.

Elasticity and the Markup

Recall Unit 1's signed demand elasticity, $\varepsilon(p) = D'(p)p/Q < 0$. At a smooth interior monopoly optimum,

$$\mathrm{MR} = p\left(1 + \frac{1}{\varepsilon}\right) = \mathrm{MC}.$$

Rearranging gives the Lerner index:

$$\boxed{\frac{p - \mathrm{MC}}{p} = \frac{1}{|\varepsilon|}.}$$

In our example,

$$|\varepsilon| = \frac{16}{6} = \frac{8}{3}, \qquad \frac{16 - 10}{16} = \frac{3}{8}.$$

More elastic demand limits the markup. With $\mathrm{MC} > 0$, an interior single-price monopoly optimum requires $|\varepsilon| > 1$.

A Monopolist Has No Independent Supply Curve

Under competition, a firm's optimal quantity can be described as a function of the price it takes as given. For a monopolist, the output choice depends on the demand curve itself:

$$P_D(Q) + QP'_D(Q) = \mathrm{MC}(Q).$$

- Both the level and slope of demand matter.
- The marginal-cost curve alone does not determine monopoly sales at a price.

A market-power counterfactual requires a model of demand and the seller's choice, as well as costs.

Policy Effects Depend on Market Structure

In our monopoly example, a ceiling of 13 allows the firm to sell up to 9 units at that price.

- Below $Q = 9$, marginal revenue is 13, which exceeds marginal cost.
- Above $Q = 9$, the original demand curve applies and marginal revenue falls below marginal cost.

The firm chooses $Q = 9$, restoring the competitive quantity in this example.

A ceiling that restricts a competitive market can expand output when it restrains market power. The benchmark matters.

A lower ceiling, operating costs, quality responses, and investment incentives can alter the conclusion.

Checkpoint: Profit and Welfare

(a) Why is MR below price for a single-price monopolist?

(b) Why does a price above marginal cost imply unrealized gains from trade in our benchmark?

(c) Does every euro of extra monopoly profit represent a euro of deadweight loss?

In Unit 5, several firms will choose while anticipating one another's responses. Monopoly provides a useful starting point for that strategic analysis.

The Welfare Benchmark Matters

Recall the result: with no external effects, competitive trade maximizes our market's total surplus at $MWTP = MC$.

Market setting	What the example establishes
Undistorted competition	A tax eliminates valuable trades. A price control can also misallocate scarce units.
Polluting competition	A corrective tax aligns $MWTP$ with social marginal cost, even though $p_b \neq p_s$.
Single-price monopoly	The seller leaves gains from trade unrealized. A suitable ceiling can expand output.

Evaluate an intervention against the actual market problem. Equal transaction prices alone do not establish efficiency, and a price wedge need not reduce welfare.

Gains from Trade and Pareto Efficiency

Return to the market **without externalities**. At $Q = 6$,

$$MWTP = 16 > MC = 10.$$

Add a small trade at 13 euros per unit, leaving earlier transactions unchanged.

Who gains?	Gain for a small quantity ΔQ
Buyer	Approximately $(16 - 13)\Delta Q$
Seller	Approximately $(13 - 10)\Delta Q$

This is a feasible Pareto improvement. At $Q = 9$, marginal value equals marginal cost.

With efficient allocation, $Q = 9$ maximizes surplus and rules out a Pareto improvement in our quasilinear benchmark. The same logic applies to exchange between goods.

Several Markets Adjust Together

So far, incomes and prices in other markets were fixed. In a broader economy, a price change can also change

- demand for substitute and complementary goods;
- the value of what households own and sell;
- production costs and incomes earned in other markets.

General equilibrium finds prices at which all individual choices are optimal and all markets clear simultaneously.

We begin with two goods and two consumers, using Unit 1's preferences and budgets.

A Pure Exchange Economy

There is no production in this example. Consumers begin with goods and can trade them.

	Good 1	Good 2	Preferences
Consumer A	8	2	$u_A = \sqrt{x_1^A x_2^A}$
Consumer B	2	8	$u_B = \sqrt{x_1^B x_2^B}$
Total resources	10	10	

$$\boldsymbol{\omega}_A = (8, 2), \qquad \boldsymbol{\omega}_B = (2, 8).$$

The endowment is what a consumer initially owns. An allocation is what each consumer ultimately consumes.

Feasibility and the Edgeworth Box

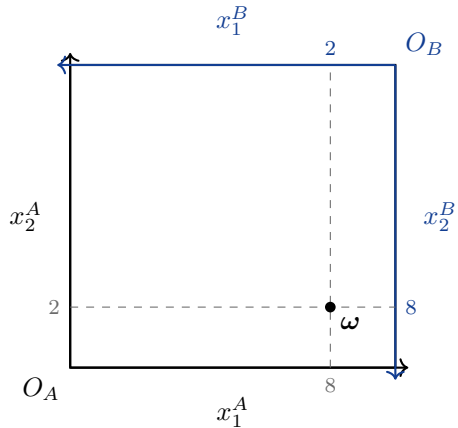
Using all resources requires

$$x_1^A + x_1^B = 10,$$

$$x_2^A + x_2^B = 10.$$

Every point in the box represents both consumers' bundles.

- Read A from the bottom left.
- Read B from the top right.



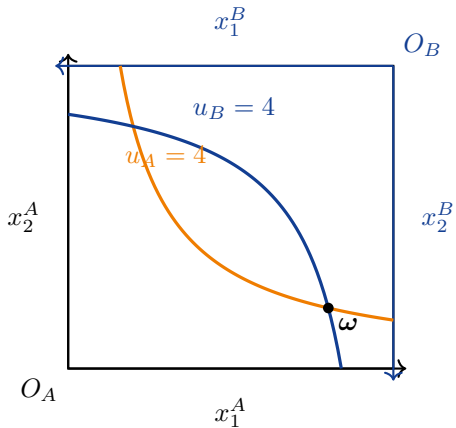
Example: A consumes $(6, 4)$ and B consumes the remaining $(4, 6)$.

Preferences in the Box

For A, more of both goods means moving toward the top right. For B, more of both goods means moving toward the bottom left. At the endowment,

$$u_A(8,2) = 4, \qquad u_B(2,8) = 4.$$

Each indifference curve collects allocations giving that consumer the same utility.



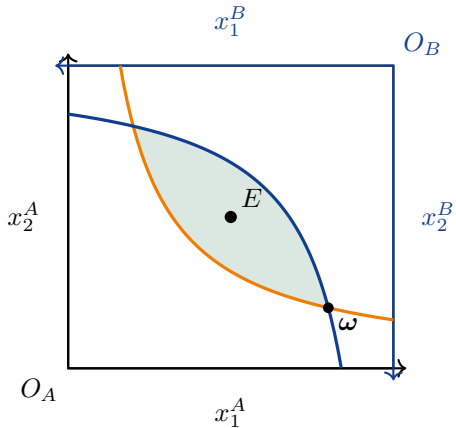
Mutually Beneficial Exchange

The shaded lens contains allocations that both consumers weakly prefer to their endowments. For example,

exchanging three units of good 1 for three of good 2 gives

$$\boldsymbol{x}_A = (5, 5), \qquad \boldsymbol{x}_B = (5, 5).$$

Both utilities rise from 4 to 5.



A Pareto improvement makes someone better off and no one worse off.

Different MRS Create Gains from Trade

Recall that MRS_{12} measures willingness to give up good 2 for a little more good 1, per additional unit of good 1. For $u_i = \sqrt{x_1^i x_2^i}$,

$$\mathrm{MRS}_{12}^i = \frac{x_2^i}{x_1^i}.$$

At the endowment,

$$\mathrm{MRS}_A = \frac{2}{8} = \frac{1}{4}, \qquad \mathrm{MRS}_B = \frac{8}{2} = 4.$$

Per marginal unit of good 1, A would accept about $1/4$ unit of good 2 and B would pay about 4. A small exchange at an intermediate rate benefits both.

Pareto Efficiency

Pareto-efficient allocation

A feasible allocation is Pareto efficient if no other feasible allocation makes at least one consumer better off without making anyone worse off.

At a smooth interior efficient allocation, gains from marginal exchanges must be exhausted:

$$\boxed{\mathrm{MRS}_A = \mathrm{MRS}_B.}$$

- Unequal MRS leave room for a mutually beneficial exchange.
- Equality gives the tangency condition for the two indifference curves.

The Contract Curve

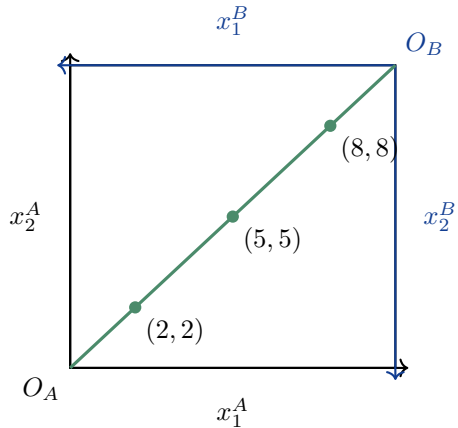
The contract curve contains the Pareto-efficient allocations. Here equal MRS and feasibility give

$$\frac{x_2^A}{x_1^A} = \frac{10 - x_2^A}{10 - x_1^A}$$

and therefore

$$\boxed{x_2^A = x_1^A.}$$

A consumes (a, a) and B consumes $(10 - a, 10 - a)$. Moving along this curve improves one consumer's utility only by reducing the other's.



Prices Turn Endowments into Budgets

At prices $p_1, p_2 > 0$, consumer i chooses a bundle satisfying

$$p_1 x_1^i + p_2 x_2^i \leq p_1 \omega_1^i + p_2 \omega_2^i.$$

Multiplying all prices by the same positive number leaves this budget set unchanged. Choose good 2 as numeraire and write $r = p_1/p_2$, so

$$\boxed{r x_1^i + x_2^i \leq r \omega_1^i + \omega_2^i = m_i(r).}$$

- The budget line passes through the endowment and has slope $-r$.
- A price change also changes the value of what the consumer owns.

Income is endogenous here. Holding purchasing power fixed after changing prices would miss part of the exchange economy.

Competitive General Equilibrium

An equilibrium consists of relative prices and a feasible allocation such that

(a) each consumer chooses a most-preferred bundle within the budget determined by those prices and the endowment;

(b) both markets clear:

$$x_1^A + x_1^B = 10, \qquad x_2^A + x_2^B = 10.$$

Optimal choices must be compatible. Two individually optimal bundles do not form an equilibrium if they require more goods than the economy has.

At an interior equilibrium, both consumers face the same price ratio:

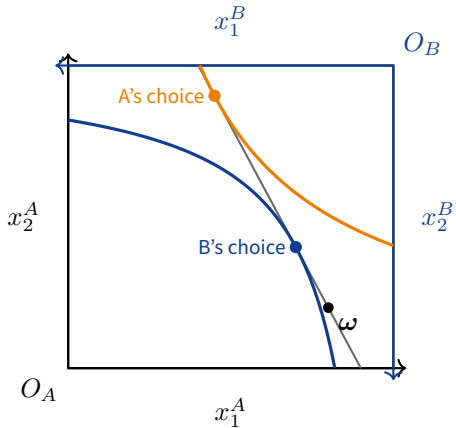
$$\mathrm{MRS}_A = r = \mathrm{MRS}_B.$$

A Price That Does Not Clear Markets

At $r = 2$, endowment incomes are $m_A = 18$ and $m_B = 12$. Each spends half on each good. A chooses $(4.5, 9)$. B chooses $(3, 6)$ from B's origin, which is the point $(7, 4)$ from A's origin. Both choices lie on the common budget line through the endowment:

$$2x_1^A + x_2^A = 18.$$

Total demand is 7.5 units of good 1 and 15 of good 2, against 10 of each available.



At equilibrium, both consumers choose the same point in the box.

The Exchange Market for Good 1

Half-budget demands for good 1 are

$$x_1^A = 4 + \frac{1}{r}, \qquad x_1^B = 1 + \frac{4}{r}.$$

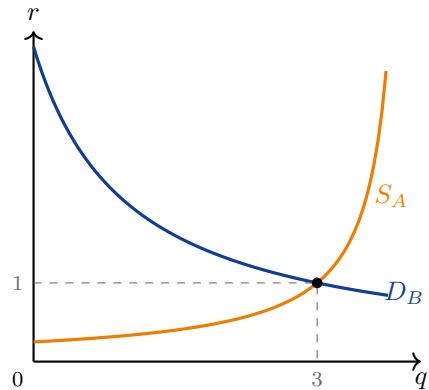
A initially owns 8, so net supply is

$$S_A(r) = 8 - x_1^A = 4 - \frac{1}{r}.$$

B initially owns 2, so net demand is

$$D_B(r) = x_1^B - 2 = \frac{4}{r} - 1.$$

At $r = 1$, A sells and B buys 3 units.



The intersection describes the same trades as the Edgeworth equilibrium. These curves include the effect of prices on endowment income.

Solving the Exchange Equilibrium

Equate net supply and demand for good 1:

$$4 - \frac{1}{r} = \frac{4}{r} - 1 \quad \Rightarrow \quad r^* = 1, \quad q^* = 3.$$

Final consumption is

$$\boxed{\boldsymbol{x}_A^* = (5,5), \qquad \boldsymbol{x}_B^* = (5,5).}$$

Check good 2:

$$x_2^A(1) + x_2^B(1) = 5 + 5 = 10.$$

A sells three units of good 1 and buys three of good 2. B makes the opposite trades. Both consumers increase utility from 4 to 5.

Equilibrium in the Edgeworth Box

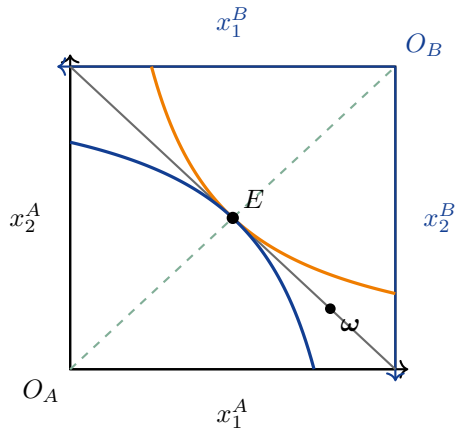
At $r = 1$, the common budget line is

$$x_1^A + x_2^A = 10.$$

It passes through the endowment and is tangent to both consumers' indifference curves at $E = (5, 5)$.

$$\mathrm{MRS}_A = 1 = \mathrm{MRS}_B.$$

The equilibrium lies on the contract curve.



The First Welfare Theorem

First Welfare Theorem

Under the standard competitive-market assumptions and local nonsatiation, every competitive equilibrium allocation is Pareto efficient.

In our smooth interior example, common prices imply

$$\mathrm{MRS}_A = \frac{p_1}{p_2} = \mathrm{MRS}_B.$$

Everyone faces the same terms of exchange, so no mutually beneficial marginal reallocation remains.

The theorem compares feasible allocations. It does not say that the resulting distribution is fair or that every policy intervention reduces welfare.

Efficiency Does Not Select a Fair Allocation

Suppose A initially owns $(8, 8)$ and B owns $(2, 2)$, with the same preferences as before. At equal prices, each already holds an optimal bundle. No trade is needed.

$$\boldsymbol{x}_A^* = (8, 8), \qquad \boldsymbol{x}_B^* = (2, 2).$$

This equilibrium is Pareto efficient, although the division of resources is unequal.

The First Welfare Theorem leaves the distribution of initial ownership open. Different endowments can support very different efficient outcomes.

The Second Welfare Theorem

Second Welfare Theorem

Under appropriate convexity and regularity conditions, a Pareto-efficient allocation can be supported as a competitive equilibrium after a suitable redistribution of wealth.

- Choose the desired efficient allocation.
- Redistribute initial purchasing power through lump-sum transfers.
- Let consumers trade at the supporting prices.

The theorem separates the distribution of purchasing power from the common marginal incentives created by prices.

Supporting a Different Efficient Allocation

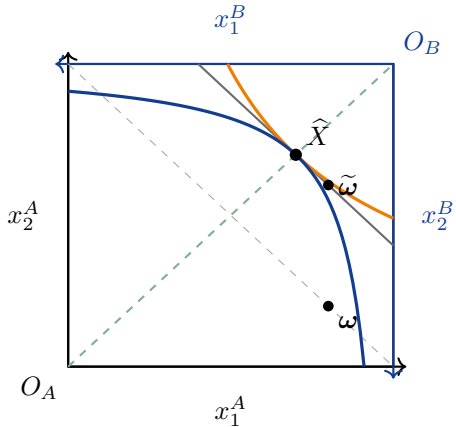
At $p_1 = p_2 = 1$, target bundles are

$$\widehat{\boldsymbol{x}}_A = (7, 7), \qquad \widehat{\boldsymbol{x}}_B = (3, 3).$$

Transfer 4 euros from B to A. Wealth changes from $(10, 10)$ to $(14, 6)$, so their half-budget choices reach the target. Equivalently, move four units of good 2 from B to A:

$$\widetilde{\boldsymbol{\omega}}_A = (8, 6), \qquad \widetilde{\boldsymbol{\omega}}_B = (2, 4).$$

A's budget becomes $x_1^A + x_2^A = 14$.



Lump-Sum Transfers and Practical Policy

A lump-sum transfer does not change with the recipient's current choices.

- It changes purchasing power while leaving the price ratio unchanged.
- Income effects can still change the chosen bundle.
- A tax on purchases or earnings changes marginal incentives.

In practice, redistribution may depend on observable choices because governments cannot observe all relevant circumstances.

> The Second Welfare Theorem describes an ideal separation of efficiency and redistribution. It does not establish that feasible redistribution is costless.

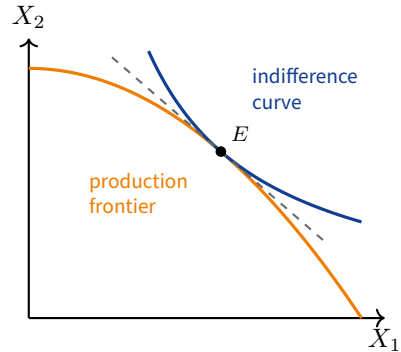
Efficiency with Production

MRS: good 2 a consumer would give up for another unit of good 1.

MRT: good 2 that production must sacrifice for another unit of good 1.

Consumers choose $MRS_i = p_1/p_2$. Efficient production maximizes output value at $MRT = p_1/p_2$.

$$\boxed{\text{MRS}_A = \text{MRS}_B = \frac{p_1}{p_2} = \text{MRT}_{12}.}$$



One-consumer illustration

With other consumption as numeraire, the frontier is $Y = \bar{Y} - C(Q)$. Its absolute slope is $MRT = C'(Q) = MC(Q)$, giving our earlier rule $MWTP = MC$.

Exercise: Endowments, Prices, and Welfare

Keep the same preferences and total resources, but let

$$\boldsymbol{\omega}_A = (9, 3), \qquad \boldsymbol{\omega}_B = (1, 7).$$

(a) Solve for the equilibrium relative price.
(b) Find each consumer's equilibrium bundle.
(c) Is the outcome efficient? Is it an equal division?
(d) What lump-sum transfers at equilibrium prices would implement $(5, 5)$ for each consumer?

Solution: Endowments and Outcomes

Aggregate demand for good 1 is still

$$\frac{9r+3}{2r}+\frac{r+7}{2r}=5+\frac{5}{r}.$$

Market clearing gives $r^*=1$. Wealth is 12 for A and 8 for B, so

$$\boldsymbol{x}_A^*=(6,6), \qquad \boldsymbol{x}_B^*=(4,4).$$

- Both bundles lie on the contract curve. The outcome is efficient.
- The division is unequal because initial wealth is unequal.
- A lump-sum transfer of 2 from A to B gives both wealth 10, implementing $(5,5)$ for each.

Redistribution changes which efficient outcome occurs. It need not make every consumer better off than before redistribution.

A Market Counterfactual

To evaluate a proposed change:

(a) Identify preferences, technology, endowments, and the initial market structure.
(b) State which incentives or constraints change.
(c) Allow the relevant choices and prices to adjust.
(d) Compute gains and losses for each affected group.
(e) Include government budgets, external effects, and allocation costs.

A credible welfare comparison combines a behavioral model, an equilibrium, and explicit choices about whose gains and losses count.

Unit 4: The Main Connections

Individual optimization	Demand and supply collect optimal choices at different prices.
Competitive equilibrium	Prices make those choices compatible.
Welfare	Gains from trade depend on quantity, allocation, and external effects.
Policy and market power	Changing incentives changes the equilibrium and the distribution of gains.
General equilibrium	Common prices coordinate several markets, with endowments shaping purchasing power.

Reading and Next Steps

Textbook reference

Ana Espinola-Arredondo and Felix Munoz-Garcia, *Intermediate Microeconomic Theory: Tools and Step-by-Step Examples*, MIT Press, 2020.

- Chapter 9: partial and general equilibrium, producer surplus, and welfare theorems.
- Chapter 10: monopoly choice, markups, and welfare.

Next unit

Strategic interactions: how choices change when firms or consumers anticipate the responses of other decision-makers.

Appendix: Two Different Consumers

Each consumer solves $\max u_i(x_i, y_i)$ subject to $px_i + y_i \leq m_i$. Assume enough income for $y_i > 0$ at the prices considered.

	Consumer A	Consumer B
Utility	$22x_A - \frac{3}{2}x_A^2 + y_A$	$22x_B - \frac{3}{4}x_B^2 + y_B$
Marginal WTP	$22 - 3x_A$	$22 - \frac{3}{2}x_B$

At an interior optimum, $MRS_i = p$, giving

$$x_A(p) = \frac{22 - p}{3}, \qquad x_B(p) = \frac{2(22 - p)}{3} \quad (0 \leq p < 22).$$

For $p \geq 22$, each chooses zero. At $p = 13$, their quantities are 3 and 6.

B's marginal willingness to pay falls more slowly with consumption. At the same price, B buys twice as much as A.

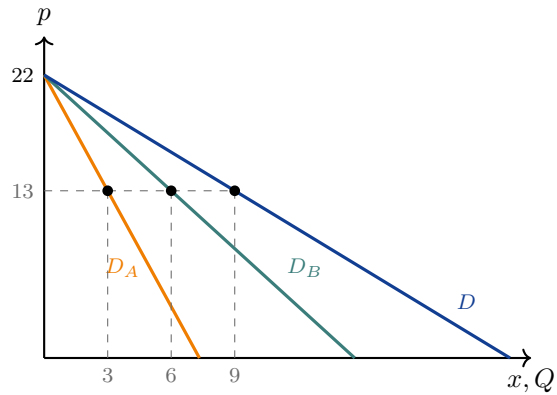
Appendix: Aggregating Demand

Add quantities at a common price:

$$\begin{aligned} D(p) &= x_A(p) + x_B(p) \\ &= \frac{22-p}{3} + \frac{2(22-p)}{3} \\ &= 22 - p. \end{aligned}$$

At $p = 13$, $D(13) = 3 + 6 = 9$.

The market curve adds *horizontal* distances. Both marginal valuations equal 13 despite different quantities.



Different individual preferences generate exactly the market demand used throughout the unit. Above $p = 22$, total demand is zero.

Appendix: Two Different Firms

Each price-taking firm solves $\max_{q_j \geq 0}\{pq_j - C_j(q_j)\}$. There are no fixed or setup costs in this example.

	Firm 1	Firm 2
Cost	$C_1(q_1) = 4q_1 + 2q_1^2$	$C_2(q_2) = 4q_2 + \frac{2}{3}q_2^2$
Marginal cost	$4 + 4q_1$	$4 + \frac{4}{3}q_2$

For $p > 4$, profit maximization requires $p = MC_j$:

$$q_1(p) = \frac{p-4}{4}, \qquad q_2(p) = \frac{3(p-4)}{4}.$$

For $p \leq 4$, both choose zero. At $p = 13$, outputs are 2.25 and 6.75.

Firm 2's marginal cost rises more slowly with output. At a common price, it produces three times as much as firm 1.

Appendix: Aggregating Supply

Add quantities at a common price:

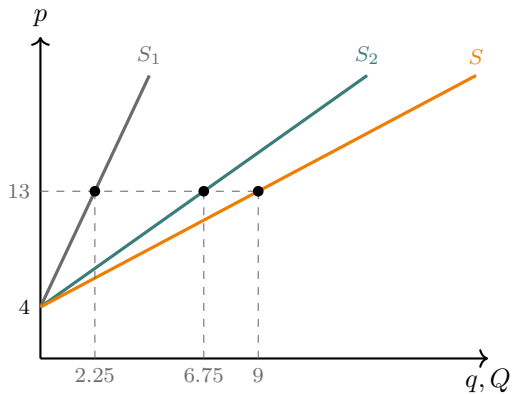
$$S(p) = q_1(p) + q_2(p) = p - 4$$

for $p > 4$. For a given total Q, cost minimization equalizes marginal costs:

$$4 + 4q_1 = 4 + \tfrac{4}{3}q_2, \qquad q_1 + q_2 = Q.$$

Thus $q_1 = Q/4$ and $q_2 = 3Q/4$. Adding the two firms' costs gives

$$\boxed{C(Q) = 4Q + \tfrac{1}{2}Q^2.}$$



The monopoly example uses these same plants and therefore the same industry cost function.

Appendix: Deriving Tax Incidence

At a taxed equilibrium,

$$D(p_b) = S(p_s), \qquad p_b - p_s = t.$$

Differentiate both conditions with respect to t:

$$D'(p_b)\frac{dp_b}{dt} = S'(p_s)\frac{dp_s}{dt}, \qquad \frac{dp_b}{dt} - \frac{dp_s}{dt} = 1.$$

Solving gives

$$\frac{dp_b}{dt} = \frac{S'}{S' - D'}, \qquad -\frac{dp_s}{dt} = \frac{-D'}{S' - D'}.$$

At an initially common price, multiplying numerator and denominator by p/Q gives the elasticity formulas in the main slides.

Appendix: Tax Size and Welfare

For our linear market, with $0 \leq t \leq 18$,

$$22 - Q_t = 4 + Q_t + t \quad \Rightarrow \quad Q_t = 9 - \frac{t}{2}.$$

The loss from the quantity reduction is

$$\text{DWL}(t) = \tfrac{1}{2} t(9 - Q_t) = t^2/4.$$

Tax	Quantity	Revenue	Deadweight loss
3	7.5	22.5	2.25
6	6	36	9
12	3	36	36

Doubling the tax quadruples deadweight loss in this linear example. Revenue need not rise proportionally because the tax base contracts.

Appendix: The Monopoly First-Order Condition

$$\pi(Q) = (22 - Q)Q - 4Q - \tfrac{1}{2}Q^2 = 18Q - \tfrac{3}{2}Q^2.$$

Thus

$$\pi'(Q) = 18 - 3Q = 0 \quad \Rightarrow \quad Q_M = 6,$$

and $\pi''(Q) = -3 < 0$ confirms the maximum. At this output,

$$R = 16 \cdot 6 = 96, \qquad C = 4 \cdot 6 + \tfrac{1}{2} \cdot 36 = 42,$$

$$\pi = \mathrm{PS} = 54.$$

Fixed operating costs would leave this interior output condition unchanged but could change whether the firm operates at all.

Appendix: Pareto Efficiency

One way to find efficient allocations is to maximize A's utility while guaranteeing B a utility level $\bar{u}_B$:

$$\max_{\boldsymbol{x}_A, \boldsymbol{x}_B} u_A(\boldsymbol{x}_A)$$

subject to

$$u_B(\boldsymbol{x}_B) \geq \bar{u}_B, \qquad x_j^A + x_j^B \leq 10 \quad (j = 1, 2).$$

At a smooth interior solution, with resource multipliers μ_1, μ_2,

$$\frac{MU_1^A}{MU_2^A} = \frac{\mu_1}{\mu_2} = \frac{MU_1^B}{MU_2^B}.$$

Changing B's guaranteed utility traces different efficient allocations. The problem does not require adding the consumers' ordinal utility numbers.

Appendix: Beyond an Exchange Economy

With production, household income can include endowments of inputs and shares in firms' profits:

$$m_i(\boldsymbol{p}) = \boldsymbol{p} \cdot \boldsymbol{\omega}_i + \sum_j \theta_{ij} \pi_j(\boldsymbol{p}).$$

- Households choose consumption and any modeled input supply.
- Firms choose feasible production plans to maximize profits.
- Every goods and input market clears.

The welfare question remains whether another feasible allocation can improve someone's welfare without harming anyone else. Production expands the set of feasible reallocations.

Appendix: Reading an Allocation

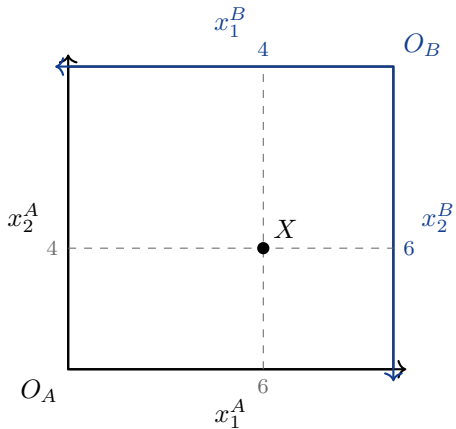
At the point $(6, 4)$ measured from A's origin,

$$\boldsymbol{x}_A = (6, 4).$$

B receives what remains:

$$\boldsymbol{x}_B = (10 - 6, 10 - 4) = (4, 6).$$

Moving right gives A more good 1 and B less good 1. Moving up gives A more good 2 and B less good 2.



Appendix: The Contract Curve

The contract curve is the set of Pareto-efficient allocations in the box. In our example, combine equal MRS with feasibility:

$$\frac{x_2^A}{x_1^A} = \frac{x_2^B}{x_1^B} = \frac{10 - x_2^A}{10 - x_1^A}.$$

Cross-multiplying gives

$$10x_2^A - x_1^A x_2^A = 10x_1^A - x_1^A x_2^A,$$

so

$$\boxed{x_2^A = x_1^A.}$$

Every efficient allocation in this example divides both goods in the same proportion between the two consumers.

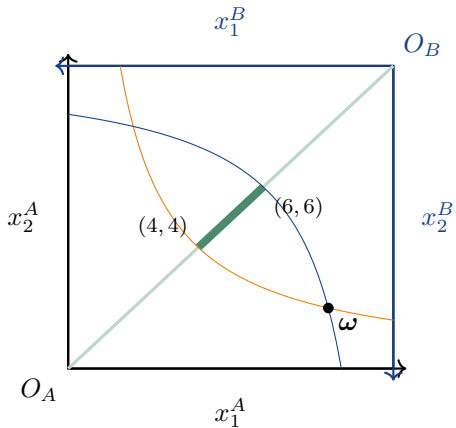
Appendix: Voluntary Exchange

The whole contract curve describes efficiency. Voluntary trade from our endowment also requires

$$u_A \geq 4, \qquad u_B \geq 4.$$

Along the curve, let A consume (a, a) and B consume $(10 - a, 10 - a)$.

$$4 \leq a \leq 6.$$



Both consumers must do at least as well as when keeping their endowments.

Appendix: Only Relative Prices Matter

Multiplying every price by the same positive number leaves all budget sets unchanged.

$$p_1 x_1 + p_2 x_2 \leq p_1 \omega_1 + p_2 \omega_2$$

is equivalent to the same inequality multiplied by $k > 0$. We therefore choose good 2 as the numeraire:

$$p_2 = 1, \qquad r = \frac{p_1}{p_2}.$$

Consumer i solves

$$\max_{x_1^i, x_2^i \geq 0} u_i(x_1^i, x_2^i) \quad \text{subject to} \quad r x_1^i + x_2^i \leq r \omega_1^i + \omega_2^i.$$

Appendix: A Trial Price

Try $r = 2$. Incomes are

$$m_A = 2(8) + 2 = 18, \qquad m_B = 2(2) + 8 = 12.$$

With the equal-share Cobb–Douglas preferences from Unit 1, each consumer spends half the budget on each good.

	Good 1	Good 2
A's demand	$18/(2 \cdot 2) = 4.5$	$18/2 = 9$
B's demand	$12/(2 \cdot 2) = 3$	$12/2 = 6$
Total demand	7.5	15
Total endowment	10	10

Good 1 has excess supply. Good 2 has excess demand.

Appendix: Individual Exchange Demands

At a general relative price r,

$$m_A = 8r + 2, \qquad m_B = 2r + 8.$$

The same optimal-choice formulas give

$$x_1^A(r) = \frac{8r+2}{2r} = 4 + \frac{1}{r}, \qquad x_2^A(r) = \frac{8r+2}{2} = 4r + 1,$$

$$x_1^B(r) = \frac{2r+8}{2r} = 1 + \frac{4}{r}, \qquad x_2^B(r) = \frac{2r+8}{2} = r + 4.$$

Every demand depends on the price ratio directly and through the value of the consumer's endowment.

Appendix: Walras's Law: Markets Are Linked

When consumers spend their full budgets, the value of aggregate excess demand is zero:

$$p_1 z_1(\boldsymbol{p}) + p_2 z_2(\boldsymbol{p}) = 0,$$

where

$$z_j = \sum_i x_j^i - \sum_i \omega_j^i.$$

At the trial price $r = 2$,

$$2(7.5 - 10) + (15 - 10) = 0.$$

In this two-good economy with positive prices, if one market clears, the other must clear too. We still verify both when checking a numerical solution.

Appendix: Why Prices Support Efficiency

At an equilibrium, each consumer has already selected a best affordable bundle.

- A strictly preferred bundle must cost more than that consumer's equilibrium budget.
- A weakly preferred bundle cannot cost less, under local nonsatiation.
- A Pareto improvement would therefore require a higher total value of consumption at the equilibrium prices.

But feasibility fixes the total resources available and their value at those prices.

A proposed Pareto improvement cannot be both affordable in aggregate and strictly better for someone.

Appendix: Supporting an Allocation

Return to the original endowments $(8, 2)$ and $(2, 8)$. Suppose the desired allocation is

$$\widehat{\boldsymbol{x}}_A = (7, 7), \qquad \widehat{\boldsymbol{x}}_B = (3, 3).$$

Both MRS equal one, so supporting prices are $p_1 = p_2 = 1$.

	Original wealth	Lump-sum transfer	New wealth
A	10	$+4$	14
B	10	-4	6

Each spends half the new wealth on each good, choosing the target allocation.

Appendix: Limits of the Welfare Theorems

- **Fairness.** Pareto efficiency does not rank all distributions.
- **Automatic adjustment.** The theorems do not describe how prices converge.
- **Universal applicability.** Market power, externalities, missing markets, and information problems can invalidate the benchmark.
- **Easy implementation.** The required lump-sum redistribution may be infeasible.

The theorems help locate the source of a market problem. They provide a benchmark for asking which assumptions fail and what an intervention changes.

Appendix: Production and Risk

With production, firms also choose inputs and output, and households earn income from inputs and ownership of firms. Efficient consumption must then be compatible with efficient production and the available resources. Unit 2 adds another dimension: households value consumption differently across uncertain states.

> Risk aversion creates potential gains from insurance. Whether markets realize those gains depends on which contracts exist and what information participants have.

Our Edgeworth example isolates exchange under certainty. The same questions about feasibility, incentives, and missing opportunities extend beyond it.